

Name of Subject :-	Machine Learning for IT Application Development	DEPT :-																																									
Sr No.	Questions	CO	Marks																																								
1	State the purpose of Machine Learning	1	2																																								
2	Explain Supervised Learning with an example.	1	2																																								
3	Where do we apply Unsupervised Learning? Write the example	1	2																																								
4	Explain the concepts of "training data" and "test data" in Machine Learning, and why they are essential.	1	2																																								
5	How the machine learning is different from traditional programming	1	2																																								
6	Differentiate Artificial Intelligence, Machine Learning and Deep Learning?	1	2																																								
7	Describe the concepts of Type I and Type II errors in hypothesis testing.	1	2																																								
8	Explain why Hypothesis Testing is important in ML	1	2																																								
9	What are some typical challenges in Machine Learning, including issues like overfitting, underfitting, and data quality problems and how can these issues be mitigated?	1	5																																								
10	Describe Bias and Variance in a Machine Learning Model	1	5																																								
11	Explain in detail, different techniques of unsupervised learning. Interpret the applications of UnSupervised Machine Learning in Modern Businesses	1	5																																								
12	Explain in detail, different techniques of Supervised learning. Interpret the applications of Supervised Machine Learning in Modern Businesses	1	5																																								
13	Contrast "null hypothesis" and "alternative hypothesis"? in hypothesis testing	1	5																																								
14	Describe different types of classification in supervised learning	1	5																																								
15	How to choose a right Machine Learning algorithm for a specific problem.	1	5																																								
16	Explain in detail different stages to develop a Machine Learning Application	1	10																																								
17	Explain different types of machine learning techniques	1	10																																								
18	Analyze various issues in machine learning	1	10																																								
19	Outline the steps involved in conducting a hypothesis test, including how to form	1	10																																								
20	Apply FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples <table><tr><th>Time</th><th>Weather</th><th>Temperature</th><th>Company</th><th>Humidity</th><th>Wind</th><th>Goes</th></tr><tr><td>Morning</td><td>Sunny</td><td>Warm</td><td>Yes</td><td>Mild</td><td>Strong</td><td>Yes</td></tr><tr><td>Evening</td><td>Rainy</td><td>Cold</td><td>No</td><td>Mild</td><td>Normal</td><td>No</td></tr><tr><td>Morning</td><td>Sunny</td><td>Moderate</td><td>Yes</td><td>Normal</td><td>Normal</td><td>Yes</td></tr><tr><td>Evening</td><td>Sunny</td><td>Cold</td><td>Yes</td><td>High</td><td>Strong</td><td>Yes</td></tr></table>	Time	Weather	Temperature	Company	Humidity	Wind	Goes	Morning	Sunny	Warm	Yes	Mild	Strong	Yes	Evening	Rainy	Cold	No	Mild	Normal	No	Morning	Sunny	Moderate	Yes	Normal	Normal	Yes	Evening	Sunny	Cold	Yes	High	Strong	Yes	1	10					
Time	Weather	Temperature	Company	Humidity	Wind	Goes																																					
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Evening	Sunny	Cold	Yes	High	Strong	Yes																																					
21	Apply FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples <table><tr><th>Example</th><th>Sky</th><th>AirTemp</th><th>Humidity</th><th>Wind</th><th>Water</th><th>Forecast</th><th>EnjoySport</th></tr><tr><td>1</td><td>Sunny</td><td>Warm</td><td>Normal</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>2</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>3</td><td>Rainy</td><td>Cold</td><td>High</td><td>Strong</td><td>Warm</td><td>Change</td><td>No</td></tr><tr><td>4</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Cool</td><td>Change</td><td>Yes</td></tr></table>	Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport	1	Sunny	Warm	Normal	Strong	Warm	Same	Yes	2	Sunny	Warm	High	Strong	Warm	Same	Yes	3	Rainy	Cold	High	Strong	Warm	Change	No	4	Sunny	Warm	High	Strong	Cool	Change	Yes	1	10
Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport																																				
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4	Sunny	Warm	High	Strong	Cool	Change	Yes																																				
22	Compare and contrast Supervised, Unsupervised, and Reinforcement Learning. Discuss the types of problems each type is best suited for, with relevant examples.	1	10																																								
23	Define feature selection and feature extraction.	2	2																																								
24	Explain why the data needs to be preprocessed?	2	2																																								
25	Describe oversampling imbalanced data handling technique.	2	2																																								

26	Define Data validation and Data Imputation	2	2																				
27	State different types of Missing values in dataset	2	2																				
28	Explain Linear discriminant analysis (LDA)	2	2																				
29	List down the reason for Data Missing from the Dataset	2	2																				
30	Discuss why data preprocessing is an important step in ML model designing	2	5																				
31	Describe the process of data normalization and its importance in Machine Learning.	2	5																				
32	Discuss the impact of data quality on Machine Learning models, including challenges like missing data and noise.	2	5																				
33	Differentiate One-Hot Encoding and Label Encoding	2	5																				
34	Analyze the process of handling outliers in a dataset and its importance in data preprocessing.	2	5																				
35	What are the advantages and disadvantages of PCA feature extraction technique?	2	5																				
36	Explain Auto encoder feature extraction technique	2	5																				
37	Explain different ways of data preprocessing in machine learning	2	10																				
38	Illustrate different feature transformation techniques used in data preprocessing	2	10																				
39	Justify “For an imbalanced class dataset F1 score is a more appropriate metric”. Also, evaluate different approaches to handle imbalanced dataset problem	2	10																				
40	Explain different types of Feature Selection Methods used in machine learning	2	10																				
41	Consider the dataset with two features (2D data). Apply PCA dimension reduction technique to modify it to one dimension. (2, 3) (3, 5) (5, 8) (8, 12) (10, 15)	2	10																				
42	Consider the dataset with two features (2D data). Apply PCA dimension reduction technique to modify it to one dimension. (3, 4) (2, 5) (5, 6) (7, 10) (10, 14)	2	10																				
43	Consider the following table and decide how chi-square test is useful to know if gender has anything to do with political party preference. Note: for degree of freedom 2 and alpha level of 0.05, tabulated value is 5.991 <table><tr><td></td><td>Republican</td><td>Democrat</td><td>Independent</td><td>Total</td></tr><tr><td>Male</td><td>100</td><td>70</td><td>30</td><td>200</td></tr><tr><td>Female</td><td>140</td><td>60</td><td>20</td><td>220</td></tr><tr><td>Total</td><td>240</td><td>130</td><td>50</td><td>440</td></tr></table>		Republican	Democrat	Independent	Total	Male	100	70	30	200	Female	140	60	20	220	Total	240	130	50	440	2	10
	Republican	Democrat	Independent	Total																			
Male	100	70	30	200																			
Female	140	60	20	220																			
Total	240	130	50	440																			
44	Describe regression in the context of machine learning	3	2																				
45	Distinguish Regression with respect to Classification	3	2																				
46	State key assumptions about simple linear regression	3	2																				
47	Explain the concept of dependent and independent variables in regression.	3	2																				
48	Evaluate polynomial regression. How does it handle non-linear relationships?	3	2																				
49	Discuss advantages and disadvantages of linear regression	3	2																				
50	Give an example of a real-world application where logistic regression is used effectively for classification	3	2																				
51	Define regularization in the context of machine learning	3	2																				
52	Interpret the role of the least squares method in linear regression	3	5																				
53	Differentiate simple linear regression and multiple linear regression techniques	3	5																				

54	The below table shows the marks obtain by student in midterm and final year exam. 1) Construct equation of predication (linear Regression formula) 2) Predict final year marks if the midterm marks is 40? <table><tr><td>Midterm marks (X)</td><td>Final exam marks (Y)</td></tr><tr><td>45</td><td>60</td></tr><tr><td>70</td><td>70</td></tr><tr><td>60</td><td>54</td></tr><tr><td>84</td><td>82</td></tr><tr><td>75</td><td>68</td></tr><tr><td>84</td><td>76</td></tr></table>	Midterm marks (X)	Final exam marks (Y)	45	60	70	70	60	54	84	82	75	68	84	76	3	5													
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45	60																													
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55	The below table shows the glucose level with respect to age of a person. 1) Construct equation of predication (linear Regression formula). 2) Predict glucose level for age 55. <table><tr><td>Age (X)</td><td>Glucose level (Y)</td></tr><tr><td>43</td><td>99</td></tr><tr><td>21</td><td>65</td></tr><tr><td>25</td><td>79</td></tr><tr><td>42</td><td>75</td></tr><tr><td>57</td><td>87</td></tr><tr><td>59</td><td>81</td></tr></table>	Age (X)	Glucose level (Y)	43	99	21	65	25	79	42	75	57	87	59	81	3	5													
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43	99																													
21	65																													
25	79																													
42	75																													
57	87																													
59	81																													
56	How can you assess the performance of a linear regression model? What evaluation metrics can be used?	3	5																											
57	Differentiate between Linear Regression and Logistic Regression. How do their output and use cases differ?	3	5																											
58	Explain the logistic function (sigmoid function) and its role in logistic regression. What is its formula, and why is it used?	3	5																											
59	Discuss key advantages of using fuzzy clustering over traditional clustering methods.	3	5																											
60	Suppose we have the following dataset with one response variable y and two predictor variables X1 and X2. Construct the final regression equation or model. <table><tr><td>y</td><td>X₁</td><td>X₂</td></tr><tr><td>140</td><td>60</td><td>22</td></tr><tr><td>155</td><td>62</td><td>25</td></tr><tr><td>159</td><td>67</td><td>24</td></tr><tr><td>179</td><td>70</td><td>20</td></tr><tr><td>192</td><td>71</td><td>15</td></tr><tr><td>200</td><td>72</td><td>14</td></tr><tr><td>212</td><td>75</td><td>14</td></tr><tr><td>215</td><td>78</td><td>11</td></tr></table>	y	X ₁	X ₂	140	60	22	155	62	25	159	67	24	179	70	20	192	71	15	200	72	14	212	75	14	215	78	11	3	10
y	X ₁	X ₂																												
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179	70	20																												
192	71	15																												
200	72	14																												
212	75	14																												
215	78	11																												
61	Discuss two popular regularization techniques, such as L1 (Lasso) and L2 (Ridge) regularization.	3	10																											

62	Consider the data points $\{(1, 3), (2, 5), (4, 8), (7, 9)\}$ along with the initial membership values for 2 clusters as given below. Revise the membership values for the given data points using fuzzy c means clustering. <table><tr><th rowspan="2">Clusters</th><th colspan="4">Points</th></tr><tr><th>(1,3)</th><th>(2,5)</th><th>(4,8)</th><th>(7,9)</th></tr><tr><td>1</td><td>0.8</td><td>0.7</td><td>0.2</td><td>0.1</td></tr><tr><td>2</td><td>0.2</td><td>0.3</td><td>0.8</td><td>0.9</td></tr></table>	Clusters	Points				(1,3)	(2,5)	(4,8)	(7,9)	1	0.8	0.7	0.2	0.1	2	0.2	0.3	0.8	0.9	3	10
Clusters	Points																					
	(1,3)	(2,5)	(4,8)	(7,9)																		
1	0.8	0.7	0.2	0.1																		
2	0.2	0.3	0.8	0.9																		
63	How will you decide which evaluation metrics to use for a particular problem, and what factors will be considered when making that choice?	3	10																			
64	The dataset of pass or fail in an exam of 5 students is given in the table. Use logistic regression as classifier to answer the following questions a) Calculate the probability of pass for the students who studied 33 hours b) At least how many hours' students should study so that he will pass the course with the probability of more than 95% Assume the model suggested by the optimizer for odds of passing the course is, $\text{Log (odds)} = -64 + 2 * \text{hours}$ <table><tr><th>Hour's study</th><th>Pass (1)/Fail (0)</th></tr><tr><td>29</td><td>0</td></tr><tr><td>15</td><td>0</td></tr><tr><td>33</td><td>1</td></tr><tr><td>28</td><td>1</td></tr><tr><td>39</td><td>1</td></tr></table>	Hour's study	Pass (1)/Fail (0)	29	0	15	0	33	1	28	1	39	1	3	10							
Hour's study	Pass (1)/Fail (0)																					
29	0																					
15	0																					
33	1																					
28	1																					
39	1																					
65	Provide an overview of the BIRCH algorithm. Apply its hierarchical clustering approach, focusing on its ability to handle large datasets by incrementally building a cluster hierarchy.	3	10																			
66	Provide an example of a real-world scenario where regression analysis can be applied effectively. Explain the specific problem domain, the variables involved, and how regression analysis can be used to make predictions.	3	10																			

Sr No.	Questions	CO	RBT	PI	Marks	Type theoretical or numerical
1	Define semi-supervised learning in the context of machine learning.	6	U	2.5.1	2	T
2	How does semi-supervised learning leverage both labeled and unlabeled data in training?	6	AN	4.4.2	2	T
3	Define reinforcement learning.	6	R	1.6.2	2	T
4	Analyze the potential risks or challenges of using semi-supervised learning in situations with limited labeled data.	6	AN	2.5.2	2	T
5	State and explain the advantages of semi-supervised learning	6	U	3.5.1	2	T
6	Provide examples of real-world applications where reinforcement learning has been successfully employed.	6	A	3.5.1	2	T
7	Describe two key benefits of using reinforcement in machine learning applications.	6	U	2.6.4	2	T
8	Discuss the challenges of semi-supervised learning in machine learning	6	AN	2.5.2	5	T
9	Provide an example of a real-world problem where semi-supervised learning would be beneficial.	6	A	3.5.1	5	T
10	Analyze the benefits of using semi-supervised learning techniques, especially in scenarios with limited labeled data. How does leveraging unlabeled data enhance the accuracy and efficiency of machine learning models?	6	AN	2.6.4	5	T
11	Discuss the application of Autonomous Vehicles and how to improve Decision-making processes for self-driving cars.	6	C	4.6.3	5	T
12	Explain the key elements of reinforcement learning, including agents, environments, actions, and rewards. How do these elements interact within the reinforcement learning framework?	6	AN	2.6.4	5	T
13	Justify with following diagram that Markov Process is the memory less random process	6	AN	2.6.4	5	N
14	For the given Markov Decision process calculate rewards for a) Start state is Class 1, move to Class 2, then class 3, then pass, then sleep. b) Start at pub and go to sleep through class 2 .	6	AN	2.6.4	5	N
15	What is Reinforcement Learning? How does it differ from other types of Machine Learning, such as Supervised Learning and unsupervised learning?	6	AN	4.6.3	10	T
16	How is Reinforcement Learning utilized in game AI? Provide an example of a popular game where Reinforcement Learning has been applied.	6	A	3.5.1	10	T

17	Explain the concept of the reward function in Reinforcement Learning. Why is it important for the learning process? Evaluate the limitations and challenges faced by reinforcement learning methods. Propose potential solutions	6	E	4.4.2	10	T
18	Explain the basic idea behind Q-learning as a Reinforcement Learning algorithm. How does it help an agent learn optimal actions?	6	E	4.4.2	10	T
19	Elaborate the application of Reinforcement Learning in robotics. How is it used to improve robotic behaviors and optimize decision-making?	6	C	2.5.2	10	T
20	Describe the benefits of using semi-supervised learning techniques, especially in scenarios with limited labeled data. How does leveraging unlabeled data enhance the accuracy and efficiency of machine learning models?	6	E	4.4.2	10	T
21	Analyze specific use cases of reinforcement learning in different domains, such as robotics, gaming, or autonomous systems. How does reinforcement learning optimize decision-making in these applications?	6	AN	2.5.2	10	T